

*Table 1. **Real-data results (Criteo)**.* AUUC on randomized test sets of increasing size (50K, 500K, 1M), for models trained on confounded training data ($n_{\text{train}} = 10\text{K}$, $n_{\text{val}} = 5\text{K}$). Results are reported mean $\pm$ std dev over five seeds. Larger test sets reduce evaluation noise in this low-outcome, low-uplift setting. Values are reported $\times 10^3$ for readability. *Higher is better*. Best mean per column is shown in **bold**.

Method	AUUC $\times 10^3$		
	50K	500K	1M
T-learner	3.58 ± 1.38	4.05 ± 1.68	4.39 ± 1.74
DR-learner	6.40 ± 1.57	7.06 ± 2.28	7.97 ± 2.31
Plug-in ranker	5.15 ± 1.86	5.18 ± 2.55	5.64 ± 3.03
Rank-learner (ours)	6.50 ± 1.26	7.30 ± 1.68	8.05 ± 1.99